

4ai	reduction in energy / amplitude (of the oscillations) due to force opposing motion / resistive forces / dissipative forces	B1 B1
4aii	amplitude is decreasing (very) <u>gradually</u> / oscillations would <u>continue</u> for a long time / many <u>complete</u> oscillations hence light damping	M1 A1
4bi	frequency = $1 / 0.3$ = 3.3 Hz	A1
4bii	energy = $\frac{1}{2} mv^2$ <u>and</u> $v = \omega x_0$ = $\frac{1}{2} \times 0.065 \times (2 \pi / 0.3)^2 \times (1.5 \times 10^{-2})^2$ = 0.00321 J = 3.2 mJ	B1 B1 A0
4c	amplitude reduces exponentially / amplitude decreases by a smaller amount (for each oscillation) / does not decrease linearly / decrease at a decreasing rate so amplitude will not be 0.7 cm	M1 A1
4d	<u>Relevant examples</u> e.g. pointer in ammeter, car suspension system etc. where <u>critical damping</u> is used	A1

5ai	Using potential divider rule	
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